MIDDLESEX SCHOOL

MATHEMATICS CURRICULUM REPORT

# Math 22

## Geometry

# Math 22 Course at a Glance

## Geometry

### Review Content

REVIEW

**LEARNING OBJECTIVES**

| | |
|---|---|
| M22-REV-001 | I can solve a linear equation arising from a geometric relationship. |
| M22-REV-002 | I can solve a proportion and interpret the resulting scale factor. |
| M22-REV-003 | I can simplify an exact square root and approximate it when needed. |
| M22-REV-004 | I can calculate distance, midpoint, and slope on the coordinate plane. |
| M22-REV-005 | I can substitute measurements into a formula with consistent units. |

### Geometry Basics

REQUIRED

**LEARNING OBJECTIVES**

| | |
|---|---|
| M22-BAS-001 | I can name and represent points, lines, planes, segments, and rays using correct notation. |
| M22-BAS-002 | I can identify collinear, coplanar, intersecting, and concurrent figures. |
| M22-BAS-003 | I can calculate segment lengths using betweenness and the segment-addition postulate. |
| M22-BAS-004 | I can determine a midpoint or use a segment bisector relationship. |
| M22-BAS-005 | I can classify and measure angles. |
| M22-BAS-006 | I can calculate angle measures using angle addition and angle bisectors. |
| M22-BAS-007 | I can identify and use complementary, supplementary, vertical, and linear-pair relationships. |
| M22-BAS-008 | I can distinguish a definition, postulate, theorem, converse, and counterexample. |
| M22-BAS-009 | I can determine angle relationships formed by parallel lines and a transversal. |
| M22-BAS-010 | I can use angle relationships to determine whether lines are parallel. |
| M22-BAS-011 | I can solve problems involving parallel, perpendicular, vertical, or horizontal lines. |
| M22-BAS-012 | I can construct a logical geometric argument from stated facts and accepted results. |

### Triangle Congruence and Proof

REQUIRED

**LEARNING OBJECTIVES**

| | |
|---|---|
| M22-CON-001 | I can classify triangles by sides and angles. |
| M22-CON-002 | I can use triangle angle-sum and exterior-angle relationships. |
| M22-CON-003 | I can write a congruence statement with corresponding vertices in the correct order. |
| M22-CON-004 | I can determine whether triangles are congruent by SAS. |
| M22-CON-005 | I can determine whether triangles are congruent by SSS. |
| M22-CON-006 | I can determine whether right triangles are congruent by HL. |
| M22-CON-007 | I can determine whether triangles are congruent by ASA or AAS. |
| M22-CON-008 | I can apply isosceles and equilateral triangle theorems and their converses. |
| M22-CON-009 | I can use corresponding parts of congruent triangles after proving congruence. |
| M22-CON-010 | I can write a structured proof involving triangle congruence. |

### Quadrilaterals, Perimeter, and Area

REQUIRED

**LEARNING OBJECTIVES**

| | |
|---|---|
| M22-QUAD-001 | I can calculate interior or exterior angle measures of a polygon. |
| M22-QUAD-002 | I can calculate perimeter and area of common polygons. |
| M22-QUAD-003 | I can apply properties of parallelograms to determine unknown measures. |
| M22-QUAD-004 | I can determine whether a quadrilateral is a parallelogram. |
| M22-QUAD-005 | I can apply properties of rectangles, rhombi, and squares. |
| M22-QUAD-006 | I can classify a quadrilateral from its stated or marked properties. |
| M22-QUAD-007 | I can apply trapezoid and trapezoid-midsegment relationships. |
| M22-QUAD-008 | I can apply properties of kites. |
| M22-QUAD-009 | I can calculate the perimeter or area of a composite polygon. |
| M22-QUAD-010 | I can justify a quadrilateral conclusion using definitions and theorems. |

# Math 22 Course at a Glance

## Geometry

### Similarity and Proportional Reasoning

REQUIRED

LEARNING OBJECTIVES

| | |
|---|---|
| M22-SIM-001 | I can determine image lengths and coordinates under a dilation. |
| M22-SIM-002 | I can identify the scale factor of a dilation or similarity relationship. |
| M22-SIM-003 | I can write a similarity statement with corresponding vertices in the correct order. |
| M22-SIM-004 | I can use proportional corresponding sides to solve similar-polygon problems. |
| M22-SIM-005 | I can relate similarity scale factor to perimeter and area scale factors. |
| M22-SIM-006 | I can prove triangles similar by AA, SSS, or SAS. |
| M22-SIM-007 | I can apply triangle proportionality theorems and their converses. |
| M22-SIM-008 | I can solve and justify an indirect-measurement or other similarity application. |

### Right Triangles

REQUIRED

LEARNING OBJECTIVES

| | |
|---|---|
| M22-RGT-001 | I can apply the Pythagorean theorem to determine a missing side. |
| M22-RGT-002 | I can use the converse of the Pythagorean theorem to classify a triangle. |
| M22-RGT-003 | I can apply triangle-inequality relationships. |
| M22-RGT-004 | I can determine exact side lengths in 45-45-90 and 30-60-90 triangles. |
| M22-RGT-005 | I can apply right-triangle similarity relationships. |
| M22-RGT-006 | I can use geometric-mean relationships created by an altitude to the hypotenuse. |
| M22-RGT-007 | I can select sine, cosine, or tangent for a right-triangle problem. |
| M22-RGT-008 | I can determine a missing side or angle of a right triangle. |
| M22-RGT-009 | I can solve a contextual right-triangle problem. |
| M22-RGT-010 | I can justify a right-triangle solution using an appropriate theorem or ratio. |

### Circles

REQUIRED

LEARNING OBJECTIVES

| | |
|---|---|
| M22-CIR-001 | I can identify and name radii, diameters, chords, secants, tangents, arcs, sectors, and segments. |
| M22-CIR-002 | I can calculate circumference and arc length. |
| M22-CIR-003 | I can calculate the area of a circle, sector, or segment. |
| M22-CIR-004 | I can determine angle and arc measures involving central and inscribed angles. |
| M22-CIR-005 | I can apply relationships among chords, arcs, and distances from the center. |
| M22-CIR-006 | I can apply tangent-radius and tangent-segment relationships. |
| M22-CIR-007 | I can determine angle measures formed by chords, secants, or tangents. |
| M22-CIR-008 | I can determine segment lengths formed by intersecting chords. |
| M22-CIR-009 | I can determine segment lengths formed by secants or tangents. |
| M22-CIR-010 | I can solve a multi-step circle problem and justify each relationship used. |

# Math 22 In-Depth Curriculum

Geometry

SKILL PROGRESSION

I Introduce | R Reinforce | D Deepen | A Apply

## Review Content

REVIEW

### LEARNING OBJECTIVES WITH SUPPORTING SKILLS

**M22-REV-001** — **I can solve a linear equation arising from a geometric relationship.**

SUPPORTING SKILLS

| | |
|---|---|
| A | Check a proposed solution in the original equation. (first introduced in Math 12)<br>SK-ALG-CHECK-SOLUTION · Apply |
| R | Use inverse operations to maintain an equivalent equation. (first introduced in Math 12)<br>SK-ALG-INVERSE-OPERATIONS · Reinforce |
| A | Translate quantities and relationships into an equation. (first introduced in Math 12)<br>SK-ALG-TRANSLATE-EQUATION · Apply |

**M22-REV-002** — **I can solve a proportion and interpret the resulting scale factor.**

SUPPORTING SKILLS

| | |
|---|---|
| R | Use cross multiplication when two ratios form a proportion. (inherited from middle school)<br>SK-ALG-CROSS-MULTIPLY · Reinforce |
| I | Calculate and interpret a similarity or dilation scale factor.<br>SK-SIM-SCALE-FACTOR · Introduce |

**M22-REV-003** — **I can simplify an exact square root and approximate it when needed.**

SUPPORTING SKILLS

| | |
|---|---|
| I | Distinguish exact values from measured or rounded approximations.<br>SK-GEO-EXACT-APPROX · Introduce |
| R | Extract perfect-power factors from a radical. (first introduced in Math 21)<br>SK-RAD-FACTOR-PERFECT · Reinforce |

**M22-REV-004** — **I can calculate distance, midpoint, and slope on the coordinate plane.**

SUPPORTING SKILLS

| | |
|---|---|
| R | Calculate horizontal or vertical distance from the difference of coordinates. (inherited horizontal/vertical coordinate distance)<br>SK-COORD-DISTANCE-AXIS · Reinforce |
| I | Calculate non-axis-aligned distance using the Pythagorean theorem or distance formula.<br>SK-COORD-DISTANCE-PYTHAGOREAN · Introduce |
| R | Calculate the midpoint of a segment from endpoint coordinates. (inherited from earlier mathematics)<br>SK-COORD-MIDPOINT · Reinforce |
| R | Calculate slope from two points. (first introduced in Math 12)<br>SK-LIN-SLOPE-POINTS · Reinforce |

**M22-REV-005** — **I can substitute measurements into a formula with consistent units.**

SUPPORTING SKILLS

| | |
|---|---|
| I | Select and report appropriate linear, square, or angular units.<br>SK-GEO-UNITS · Introduce |
| A | Apply the order of operations to a numerical expression. (inherited from middle school)<br>SK-NUM-ORDER-OPERATIONS · Apply |

# Math 22 In-Depth Curriculum

Geometry

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

## Geometry Basics

REQUIRED

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

### M22-BAS-001 — I can name and represent points, lines, planes, segments, and rays using correct notation.

SUPPORTING SKILLS

| | |
|---|---|
| I | Add congruence, parallel, perpendicular, and angle markings to a diagram.<br>SK-GEO-DIAGRAM-MARK · Introduce |
| I | Read and write geometric notation correctly.<br>SK-GEO-NOTATION · Introduce |

### M22-BAS-002 — I can identify collinear, coplanar, intersecting, and concurrent figures.

SUPPORTING SKILLS

| | |
|---|---|
| I | Distinguish stated or marked facts from visual appearance.<br>SK-GEO-DIAGRAM-INFER · Introduce |
| A | Read and write geometric notation correctly. (first introduced in Math 22)<br>SK-GEO-NOTATION · Apply |

### M22-BAS-003 — I can calculate segment lengths using betweenness and the segment-addition postulate.

SUPPORTING SKILLS

| | |
|---|---|
| A | Use inverse operations to maintain an equivalent equation. (first introduced in Math 12)<br>SK-ALG-INVERSE-OPERATIONS · Apply |
| A | Translate quantities and relationships into an equation. (first introduced in Math 12)<br>SK-ALG-TRANSLATE-EQUATION · Apply |
| I | Translate segment addition into an equation.<br>SK-GEO-SEGMENT-ADDITION · Introduce |

### M22-BAS-004 — I can determine a midpoint or use a segment bisector relationship.

SUPPORTING SKILLS

| | |
|---|---|
| I | Use a bisector to create equal measures.<br>SK-GEO-BISECTOR · Introduce |
| I | Use a midpoint to create equal segment lengths.<br>SK-GEO-MIDPOINT · Introduce |
| A | Translate segment addition into an equation. (first introduced in Math 22)<br>SK-GEO-SEGMENT-ADDITION · Apply |

### M22-BAS-005 — I can classify and measure angles.

SUPPORTING SKILLS

| | |
|---|---|
| I | Identify complementary, supplementary, vertical, and linear-pair relationships.<br>SK-ANG-PAIR-IDENTIFY · Introduce |
| I | Measure or calculate an angle accurately.<br>SK-GEO-ANGLE-MEASURE · Introduce |

### M22-BAS-006 — I can calculate angle measures using angle addition and angle bisectors.

SUPPORTING SKILLS

| | |
|---|---|
| A | Use inverse operations to maintain an equivalent equation. (first introduced in Math 12)<br>SK-ALG-INVERSE-OPERATIONS · Apply |
| I | Translate angle addition into an equation.<br>SK-GEO-ANGLE-ADDITION · Introduce |
| A | Use a bisector to create equal measures. (first introduced in Math 22)<br>SK-GEO-BISECTOR · Apply |

# Math 22 In-Depth Curriculum

## Geometry

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

### M22-BAS-007 — I can identify and use complementary, supplementary, vertical, and linear-pair relationships.

SUPPORTING SKILLS

| | |
|---|---|
| A | Use inverse operations to maintain an equivalent equation. (first introduced in Math 12)<br>SK-ALG-INVERSE-OPERATIONS · Apply |
| A | Translate quantities and relationships into an equation. (first introduced in Math 12)<br>SK-ALG-TRANSLATE-EQUATION · Apply |
| D | Identify complementary, supplementary, vertical, and linear-pair relationships. (first introduced in Math 22)<br>SK-ANG-PAIR-IDENTIFY · Deepen |

### M22-BAS-008 — I can distinguish a definition, postulate, theorem, converse, and counterexample.

SUPPORTING SKILLS

| | |
|---|---|
| I | Recognize and apply the converse of a conditional statement.<br>SK-PROOF-CONVERSE · Introduce |
| I | Disprove a universal statement with a counterexample.<br>SK-PROOF-COUNTEREXAMPLE · Introduce |
| I | Apply a geometric definition as justification.<br>SK-PROOF-DEFINITION · Introduce |
| I | Select a theorem that connects known facts to a target statement.<br>SK-PROOF-THEOREM-SELECT · Introduce |

### M22-BAS-009 — I can determine angle relationships formed by parallel lines and a transversal.

SUPPORTING SKILLS

| | |
|---|---|
| I | Use parallel-line angle relationships.<br>SK-ANG-PARALLEL-APPLY · Introduce |
| I | Identify corresponding, alternate, and same-side angle pairs.<br>SK-ANG-TRANSVERSAL-IDENTIFY · Introduce |

### M22-BAS-010 — I can use angle relationships to determine whether lines are parallel.

SUPPORTING SKILLS

| | |
|---|---|
| I | Use angle relationships to prove lines parallel.<br>SK-ANG-PARALLEL-CONVERSE · Introduce |
| A | Identify corresponding, alternate, and same-side angle pairs. (first introduced in Math 22)<br>SK-ANG-TRANSVERSAL-IDENTIFY · Apply |

### M22-BAS-011 — I can solve problems involving parallel, perpendicular, vertical, or horizontal lines.

SUPPORTING SKILLS

| | |
|---|---|
| A | Use inverse operations to maintain an equivalent equation. (first introduced in Math 12)<br>SK-ALG-INVERSE-OPERATIONS · Apply |
| A | Use parallel-line angle relationships. (first introduced in Math 22)<br>SK-ANG-PARALLEL-APPLY · Apply |
| I | Use right-angle relationships created by perpendicular lines.<br>SK-LIN-PERPENDICULAR-ANGLE · Introduce |

### M22-BAS-012 — I can construct a logical geometric argument from stated facts and accepted results.

SUPPORTING SKILLS

| | |
|---|---|
| I | Translate givens into diagram markings and statements.<br>SK-PROOF-GIVEN · Introduce |
| I | Organize statements and reasons into a valid proof sequence.<br>SK-PROOF-STRUCTURE · Introduce |
| A | Select a theorem that connects known facts to a target statement. (first introduced in Math 22)<br>SK-PROOF-THEOREM-SELECT · Apply |

# Math 22 In-Depth Curriculum

Geometry

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

## Triangle Congruence and Proof

REQUIRED

### LEARNING OBJECTIVES WITH SUPPORTING SKILLS

**M22-CON-001** — **I can classify triangles by sides and angles.**

SUPPORTING SKILLS

| | |
|---|---|
| A | Add congruence, parallel, perpendicular, and angle markings to a diagram. (first introduced in Math 22)<br>SK-GEO-DIAGRAM-MARK · Apply |
| I | Classify a triangle by side lengths and angle measures.<br>SK-TRI-CLASSIFY · Introduce |

**M22-CON-002** — **I can use triangle angle-sum and exterior-angle relationships.**

SUPPORTING SKILLS

| | |
|---|---|
| A | Use inverse operations to maintain an equivalent equation. (first introduced in Math 12)<br>SK-ALG-INVERSE-OPERATIONS · Apply |
| I | Apply the triangle angle-sum theorem.<br>SK-TRI-ANGLE-SUM · Introduce |
| I | Apply the triangle exterior-angle theorem.<br>SK-TRI-EXTERIOR-ANGLE · Introduce |

**M22-CON-003** — **I can write a congruence statement with corresponding vertices in the correct order.**

SUPPORTING SKILLS

| | |
|---|---|
| I | Match corresponding vertices, sides, and angles in order.<br>SK-GEO-CORRESPONDENCE · Introduce |
| I | Write congruence statements in corresponding order.<br>SK-PROOF-CONGRUENCE-ORDER · Introduce |

**M22-CON-004** — **I can determine whether triangles are congruent by SAS.**

SUPPORTING SKILLS

| | |
|---|---|
| A | Match corresponding vertices, sides, and angles in order. (first introduced in Math 22)<br>SK-GEO-CORRESPONDENCE · Apply |
| D | Select a theorem that connects known facts to a target statement. (first introduced in Math 22)<br>SK-PROOF-THEOREM-SELECT · Deepen |
| I | Select SAS, SSS, HL, ASA, or AAS from marked information.<br>SK-TRI-CONGRUENCE-SELECT · Introduce |

**M22-CON-005** — **I can determine whether triangles are congruent by SSS.**

SUPPORTING SKILLS

| | |
|---|---|
| A | Match corresponding vertices, sides, and angles in order. (first introduced in Math 22)<br>SK-GEO-CORRESPONDENCE · Apply |
| D | Select SAS, SSS, HL, ASA, or AAS from marked information. (first introduced in Math 22)<br>SK-TRI-CONGRUENCE-SELECT · Deepen |

**M22-CON-006** — **I can determine whether right triangles are congruent by HL.**

SUPPORTING SKILLS

| | |
|---|---|
| A | Use right-angle relationships created by perpendicular lines. (first introduced in Math 22)<br>SK-LIN-PERPENDICULAR-ANGLE · Apply |
| D | Select SAS, SSS, HL, ASA, or AAS from marked information. (first introduced in Math 22)<br>SK-TRI-CONGRUENCE-SELECT · Deepen |

**M22-CON-007** — **I can determine whether triangles are congruent by ASA or AAS.**

SUPPORTING SKILLS

| | |
|---|---|
| A | Apply the triangle angle-sum theorem. (first introduced in Math 22)<br>SK-TRI-ANGLE-SUM · Apply |
| D | Select SAS, SSS, HL, ASA, or AAS from marked information. (first introduced in Math 22)<br>SK-TRI-CONGRUENCE-SELECT · Deepen |

# Math 22 In-Depth Curriculum

## Geometry

SKILL PROGRESSION

I Introduce | R Reinforce | D Deepen | A Apply

### M22-CON-008 — I can apply isosceles and equilateral triangle theorems and their converses.

SUPPORTING SKILLS

| | |
|---|---|
| D | Recognize and apply the converse of a conditional statement. (first introduced in Math 22)<br>SK-PROOF-CONVERSE · Deepen |
| I | Apply isosceles-triangle theorems and converses.<br>SK-TRI-ISOSCELES · Introduce |

### M22-CON-009 — I can use corresponding parts of congruent triangles after proving congruence.

SUPPORTING SKILLS

| | |
|---|---|
| A | Write congruence statements in corresponding order. (first introduced in Math 22)<br>SK-PROOF-CONGRUENCE-ORDER · Apply |
| I | Use corresponding parts only after triangle congruence is established.<br>SK-PROOF-CPCTC · Introduce |

### M22-CON-010 — I can write a structured proof involving triangle congruence.

SUPPORTING SKILLS

| | |
|---|---|
| A | Use corresponding parts only after triangle congruence is established. (first introduced in Math 22)<br>SK-PROOF-CPCTC · Apply |
| A | Translate givens into diagram markings and statements. (first introduced in Math 22)<br>SK-PROOF-GIVEN · Apply |
| D | Organize statements and reasons into a valid proof sequence. (first introduced in Math 22)<br>SK-PROOF-STRUCTURE · Deepen |
| A | Select SAS, SSS, HL, ASA, or AAS from marked information. (first introduced in Math 22)<br>SK-TRI-CONGRUENCE-SELECT · Apply |

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

# Quadrilaterals, Perimeter, and Area

REQUIRED

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

### M22-QUA D-001 — I can calculate interior or exterior angle measures of a polygon.

SUPPORTING SKILLS

| | |
|---|---|
| A | Use inverse operations to maintain an equivalent equation. (first introduced in Math 12)<br>SK-ALG-INVERSE-OPERATIONS · Apply |
| I | Calculate polygon interior or exterior angle sums.<br>SK-POLY-ANGLE-SUM · Introduce |

### M22-QUA D-002 — I can calculate perimeter and area of common polygons.

SUPPORTING SKILLS

| | |
|---|---|
| I | Add boundary lengths without omitting or duplicating sides.<br>SK-GEO-PERIMETER · Introduce |
| D | Select and report appropriate linear, square, or angular units. (first introduced in Math 22)<br>SK-GEO-UNITS · Deepen |
| A | Apply the order of operations to a numerical expression. (inherited from middle school)<br>SK-NUM-ORDER-OPERATIONS · Apply |

### M22-QUA D-003 — I can apply properties of parallelograms to determine unknown measures.

SUPPORTING SKILLS

| | |
|---|---|
| A | Translate quantities and relationships into an equation. (first introduced in Math 12)<br>SK-ALG-TRANSLATE-EQUATION · Apply |
| I | Apply parallelogram properties and tests.<br>SK-QUAD-PARALLELOGRAM · Introduce |

### M22-QUA D-004 — I can determine whether a quadrilateral is a parallelogram.

SUPPORTING SKILLS

| | |
|---|---|
| A | Recognize and apply the converse of a conditional statement. (first introduced in Math 22)<br>SK-PROOF-CONVERSE · Apply |
| D | Apply parallelogram properties and tests. (first introduced in Math 22)<br>SK-QUAD-PARALLELOGRAM · Deepen |

### M22-QUA D-005 — I can apply properties of rectangles, rhombi, and squares.

SUPPORTING SKILLS

| | |
|---|---|
| I | Use the inclusive hierarchy of quadrilateral classifications.<br>SK-QUAD-HIERARCHY · Introduce |
| I | Distinguish and apply rectangle, rhombus, and square properties.<br>SK-QUAD-RECT-RHOM-SQUARE · Introduce |

### M22-QUA D-006 — I can classify a quadrilateral from its stated or marked properties.

SUPPORTING SKILLS

| | |
|---|---|
| A | Distinguish stated or marked facts from visual appearance. (first introduced in Math 22)<br>SK-GEO-DIAGRAM-INFER · Apply |
| D | Use the inclusive hierarchy of quadrilateral classifications. (first introduced in Math 22)<br>SK-QUAD-HIERARCHY · Deepen |

### M22-QUA D-007 — I can apply trapezoid and trapezoid-midsegment relationships.

SUPPORTING SKILLS

| | |
|---|---|
| A | Use inverse operations to maintain an equivalent equation. (first introduced in Math 12)<br>SK-ALG-INVERSE-OPERATIONS · Apply |
| I | Apply trapezoid and midsegment relationships.<br>SK-QUAD-TRAPEZOID · Introduce |

# Math 22 In-Depth Curriculum

## Geometry

SKILL PROGRESSION

| | |
|---|---|
| I | Introduce |
| R | Reinforce |
| D | Deepen |
| A | Apply |

### M22-QUAD-008 — I can apply properties of kites.

SUPPORTING SKILLS

| | |
|---|---|
| I | Apply kite properties.<br>SK-QUAD-KITE · Introduce |

### M22-QUAD-009 — I can calculate the perimeter or area of a composite polygon.

SUPPORTING SKILLS

| | |
|---|---|
| I | Decompose a composite region into familiar shapes.<br>SK-GEO-AREA-DECOMPOSE · Introduce |
| D | Add boundary lengths without omitting or duplicating sides. (first introduced in Math 22)<br>SK-GEO-PERIMETER · Deepen |
| A | Select and report appropriate linear, square, or angular units. (first introduced in Math 22)<br>SK-GEO-UNITS · Apply |

### M22-QUAD-010 — I can justify a quadrilateral conclusion using definitions and theorems.

SUPPORTING SKILLS

| | |
|---|---|
| D | Apply a geometric definition as justification. (first introduced in Math 22)<br>SK-PROOF-DEFINITION · Deepen |
| A | Organize statements and reasons into a valid proof sequence. (first introduced in Math 22)<br>SK-PROOF-STRUCTURE · Apply |
| D | Select a theorem that connects known facts to a target statement. (first introduced in Math 22)<br>SK-PROOF-THEOREM-SELECT · Deepen |

SKILL PROGRESSION

I Introduce | R Reinforce | D Deepen | A Apply

# Similarity and Proportional Reasoning

REQUIRED

## LEARNING OBJECTIVES WITH SUPPORTING SKILLS

### M22-SIM-01 — I can determine image lengths and coordinates under a dilation.

SUPPORTING SKILLS

| | |
|---|---|
| A | Match corresponding vertices, sides, and angles in order. (first introduced in Math 22)<br>SK-GEO-CORRESPONDENCE · Apply |
| D | Calculate and interpret a similarity or dilation scale factor. (first introduced in Math 22)<br>SK-SIM-SCALE-FACTOR · Deepen |

### M22-SIM-02 — I can identify the scale factor of a dilation or similarity relationship.

SUPPORTING SKILLS

| | |
|---|---|
| A | Use cross multiplication when two ratios form a proportion. (inherited from middle school)<br>SK-ALG-CROSS-MULTIPLY · Apply |
| D | Calculate and interpret a similarity or dilation scale factor. (first introduced in Math 22)<br>SK-SIM-SCALE-FACTOR · Deepen |

### M22-SIM-03 — I can write a similarity statement with corresponding vertices in the correct order.

SUPPORTING SKILLS

| | |
|---|---|
| D | Match corresponding vertices, sides, and angles in order. (first introduced in Math 22)<br>SK-GEO-CORRESPONDENCE · Deepen |
| I | Write similarity statements in corresponding order.<br>SK-PROOF-SIMILARITY-ORDER · Introduce |

### M22-SIM-04 — I can use proportional corresponding sides to solve similar-polygon problems.

SUPPORTING SKILLS

| | |
|---|---|
| A | Use cross multiplication when two ratios form a proportion. (inherited from middle school)<br>SK-ALG-CROSS-MULTIPLY · Apply |
| A | Match corresponding vertices, sides, and angles in order. (first introduced in Math 22)<br>SK-GEO-CORRESPONDENCE · Apply |
| I | Set up proportions using corresponding sides.<br>SK-SIM-PROPORTION · Introduce |

### M22-SIM-05 — I can relate similarity scale factor to perimeter and area scale factors.

SUPPORTING SKILLS

| | |
|---|---|
| I | Relate length scale factor to perimeter and area factors.<br>SK-SIM-PERIMETER-AREA · Introduce |
| A | Calculate and interpret a similarity or dilation scale factor. (first introduced in Math 22)<br>SK-SIM-SCALE-FACTOR · Apply |

### M22-SIM-06 — I can prove triangles similar by AA, SSS, or SAS.

SUPPORTING SKILLS

| | |
|---|---|
| A | Write similarity statements in corresponding order. (first introduced in Math 22)<br>SK-PROOF-SIMILARITY-ORDER · Apply |
| D | Organize statements and reasons into a valid proof sequence. (first introduced in Math 22)<br>SK-PROOF-STRUCTURE · Deepen |
| I | Recognize AA triangle similarity.<br>SK-SIM-AA · Introduce |
| I | Test SSS or SAS triangle similarity.<br>SK-SIM-SSS-SAS · Introduce |

# Math 22 In-Depth Curriculum

## Geometry

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

### M22-SIM-007 — I can apply triangle proportionality theorems and their converses.

SUPPORTING SKILLS

| | |
|---|---|
| A | Recognize and apply the converse of a conditional statement. (first introduced in Math 22)<br>SK-PROOF-CONVERSE · Apply |
| I | Apply triangle proportionality relationships.<br>SK-SIM-TRI-PROPORTIONALITY · Introduce |

### M22-SIM-008 — I can solve and justify an indirect-measurement or other similarity application.

SUPPORTING SKILLS

| | |
|---|---|
| A | Select a theorem that connects known facts to a target statement. (first introduced in Math 22)<br>SK-PROOF-THEOREM-SELECT · Apply |
| D | Set up proportions using corresponding sides. (first introduced in Math 22)<br>SK-SIM-PROPORTION · Deepen |
| A | Calculate and interpret a similarity or dilation scale factor. (first introduced in Math 22)<br>SK-SIM-SCALE-FACTOR · Apply |

# Math 22 In-Depth Curriculum

Geometry

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

## Right Triangles

REQUIRED

### LEARNING OBJECTIVES WITH SUPPORTING SKILLS

#### M22-RGT-001 I can apply the Pythagorean theorem to determine a missing side.

SUPPORTING SKILLS

| | |
|---|---|
| A | Extract perfect-power factors from a radical. (first introduced in Math 21)<br>SK-RAD-FACTOR-PERFECT · Apply |
| I | Apply the Pythagorean theorem or its converse.<br>SK-TRI-PYTHAGOREAN · Introduce |

#### M22-RGT-002 I can use the converse of the Pythagorean theorem to classify a triangle.

SUPPORTING SKILLS

| | |
|---|---|
| A | Recognize and apply the converse of a conditional statement. (first introduced in Math 22)<br>SK-PROOF-CONVERSE · Apply |
| D | Apply the Pythagorean theorem or its converse. (first introduced in Math 22)<br>SK-TRI-PYTHAGOREAN · Deepen |

#### M22-RGT-003 I can apply triangle-inequality relationships.

SUPPORTING SKILLS

| | |
|---|---|
| A | Select a theorem that connects known facts to a target statement. (first introduced in Math 22)<br>SK-PROOF-THEOREM-SELECT · Apply |
| A | Classify a triangle by side lengths and angle measures. (first introduced in Math 22)<br>SK-TRI-CLASSIFY · Apply |

#### M22-RGT-004 I can determine exact side lengths in 45-45-90 and 30-60-90 triangles.

SUPPORTING SKILLS

| | |
|---|---|
| A | Distinguish exact values from measured or rounded approximations. (first introduced in Math 22)<br>SK-GEO-EXACT-APPROX · Apply |
| R | Extract perfect-power factors from a radical. (first introduced in Math 21)<br>SK-RAD-FACTOR-PERFECT · Reinforce |
| I | Recall special-right-triangle ratios.<br>SK-TRI-SPECIAL-RATIOS · Introduce |

#### M22-RGT-005 I can apply right-triangle similarity relationships.

SUPPORTING SKILLS

| | |
|---|---|
| A | Recognize AA triangle similarity. (first introduced in Math 22)<br>SK-SIM-AA · Apply |
| D | Set up proportions using corresponding sides. (first introduced in Math 22)<br>SK-SIM-PROPORTION · Deepen |

#### M22-RGT-006 I can use geometric-mean relationships created by an altitude to the hypotenuse.

SUPPORTING SKILLS

| | |
|---|---|
| A | Set up proportions using corresponding sides. (first introduced in Math 22)<br>SK-SIM-PROPORTION · Apply |
| I | Apply geometric-mean relationships in a right triangle split by an altitude.<br>SK-TRI-GEOMETRIC-MEAN · Introduce |

# Math 22 In-Depth Curriculum

## Geometry

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### M22-RGT-007 — I can select sine, cosine, or tangent for a right-triangle problem.

SUPPORTING SKILLS

| | |
|---|---|
| A | Add congruence, parallel, perpendicular, and angle markings to a diagram. (first introduced in Math 22)<br>SK-GEO-DIAGRAM-MARK · Apply |
| I | Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio.<br>SK-TRI-TRIG-RATIO · Introduce |
| I | Label sides relative to a selected angle.<br>SK-TRIG-TRIANGLE-LABEL · Introduce |

### M22-RGT-008 — I can determine a missing side or angle of a right triangle.

SUPPORTING SKILLS

| | |
|---|---|
| A | Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio. (first introduced in Math 22)<br>SK-TRI-TRIG-RATIO · Apply |
| I | Use an inverse trigonometric function to determine a missing angle.<br>SK-TRI-TRIG-SOLVE-ANGLE · Introduce |
| I | Use a trigonometric ratio to determine a missing side.<br>SK-TRI-TRIG-SOLVE-SIDE · Introduce |

### M22-RGT-009 — I can solve a contextual right-triangle problem.

SUPPORTING SKILLS

| | |
|---|---|
| A | Restrict a model to meaningful contextual inputs. (first introduced in Math 12)<br>SK-MODEL-VALID-DOMAIN · Apply |
| D | Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio. (first introduced in Math 22)<br>SK-TRI-TRIG-RATIO · Deepen |
| A | Use an inverse trigonometric function to determine a missing angle. (first introduced in Math 22)<br>SK-TRI-TRIG-SOLVE-ANGLE · Apply |
| A | Use a trigonometric ratio to determine a missing side. (first introduced in Math 22)<br>SK-TRI-TRIG-SOLVE-SIDE · Apply |

### M22-RGT-010 — I can justify a right-triangle solution using an appropriate theorem or ratio.

SUPPORTING SKILLS

| | |
|---|---|
| D | Select a theorem that connects known facts to a target statement. (first introduced in Math 22)<br>SK-PROOF-THEOREM-SELECT · Deepen |
| A | Apply the Pythagorean theorem or its converse. (first introduced in Math 22)<br>SK-TRI-PYTHAGOREAN · Apply |
| A | Recall special-right-triangle ratios. (first introduced in Math 22)<br>SK-TRI-SPECIAL-RATIOS · Apply |
| A | Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio. (first introduced in Math 22)<br>SK-TRI-TRIG-RATIO · Apply |

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Geometry

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## Circles

REQUIRED

### LEARNING OBJECTIVES WITH SUPPORTING SKILLS

**M22-CIR-001** — **I can identify and name radii, diameters, chords, secants, tangents, arcs, sectors, and segments.**

SUPPORTING SKILLS

| | |
|---|---|
| I | Identify circle parts and relationships from a diagram.<br>SK-CIR-VOCAB · Introduce |
| A | Read and write geometric notation correctly. (first introduced in Math 22)<br>SK-GEO-NOTATION · Apply |

**M22-CIR-002** — **I can calculate circumference and arc length.**

SUPPORTING SKILLS

| | |
|---|---|
| I | Relate a central angle to a fraction of a circle.<br>SK-CIR-ARC-FRACTION · Introduce |
| I | Calculate arc length from radius and central angle.<br>SK-CIR-ARC-LENGTH · Introduce |
| I | Calculate circumference from radius or diameter.<br>SK-CIR-CIRCUMFERENCE · Introduce |
| A | Select and report appropriate linear, square, or angular units. (first introduced in Math 22)<br>SK-GEO-UNITS · Apply |

**M22-CIR-003** — **I can calculate the area of a circle, sector, or segment.**

SUPPORTING SKILLS

| | |
|---|---|
| I | Calculate the area of a circle.<br>SK-CIR-AREA · Introduce |
| I | Calculate sector area from radius and central angle.<br>SK-CIR-SECTOR-AREA · Introduce |
| I | Calculate segment area by subtracting a triangle from a sector.<br>SK-CIR-SEGMENT-AREA · Introduce |
| A | Select and report appropriate linear, square, or angular units. (first introduced in Math 22)<br>SK-GEO-UNITS · Apply |

**M22-CIR-004** — **I can determine angle and arc measures involving central and inscribed angles.**

SUPPORTING SKILLS

| | |
|---|---|
| A | Use inverse operations to maintain an equivalent equation. (first introduced in Math 12)<br>SK-ALG-INVERSE-OPERATIONS · Apply |
| D | Relate a central angle to a fraction of a circle. (first introduced in Math 22)<br>SK-CIR-ARC-FRACTION · Deepen |
| I | Relate an inscribed angle to its intercepted arc.<br>SK-CIR-INSCRIBED-ANGLE · Introduce |

**M22-CIR-005** — **I can apply relationships among chords, arcs, and distances from the center.**

SUPPORTING SKILLS

| | |
|---|---|
| I | Apply equal-chord, equal-arc, and center-distance relationships.<br>SK-CIR-CHORD-RELATION · Introduce |
| A | Identify circle parts and relationships from a diagram. (first introduced in Math 22)<br>SK-CIR-VOCAB · Apply |

**M22-CIR-006** — **I can apply tangent-radius and tangent-segment relationships.**

SUPPORTING SKILLS

| | |
|---|---|
| I | Select and apply an intersecting-segment product relationship.<br>SK-CIR-SEGMENT-PRODUCT · Introduce |
| I | Use the perpendicular relationship between a radius and tangent.<br>SK-CIR-TANGENT-RADIUS · Introduce |

**M22-CIR-007** — **I can determine angle measures formed by chords, secants, or tangents.**

SUPPORTING SKILLS

| | |
|---|---|
| I | Select the correct angle formula for chords, secants, or tangents.<br>SK-CIR-ANGLE-RELATION · Introduce |
| A | Identify circle parts and relationships from a diagram. (first introduced in Math 22)<br>SK-CIR-VOCAB · Apply |

# Math 22 In-Depth Curriculum

## Geometry

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### M22-CIR-008 I can determine segment lengths formed by intersecting chords.

SUPPORTING SKILLS

| | |
|---|---|
| A | Use inverse operations to maintain an equivalent equation. (first introduced in Math 12)<br>SK-ALG-INVERSE-OPERATIONS · Apply |
| I | Select and apply an intersecting-segment product relationship.<br>SK-CIR-SEGMENT-PRODUCT · Introduce |

### M22-CIR-009 I can determine segment lengths formed by secants or tangents.

SUPPORTING SKILLS

| | |
|---|---|
| A | Use inverse operations to maintain an equivalent equation. (first introduced in Math 12)<br>SK-ALG-INVERSE-OPERATIONS · Apply |
| D | Select and apply an intersecting-segment product relationship. (first introduced in Math 22)<br>SK-CIR-SEGMENT-PRODUCT · Deepen |

### M22-CIR-010 I can solve a multi-step circle problem and justify each relationship used.

SUPPORTING SKILLS

| | |
|---|---|
| A | Select the correct angle formula for chords, secants, or tangents. (first introduced in Math 22)<br>SK-CIR-ANGLE-RELATION · Apply |
| A | Select and apply an intersecting-segment product relationship. (first introduced in Math 22)<br>SK-CIR-SEGMENT-PRODUCT · Apply |
| A | Identify circle parts and relationships from a diagram. (first introduced in Math 22)<br>SK-CIR-VOCAB · Apply |
| D | Select a theorem that connects known facts to a target statement. (first introduced in Math 22)<br>SK-PROOF-THEOREM-SELECT · Deepen |